

Team TheParents

Project AMI

Status Report 3

Presenter: Stefano Buglione, SAB21S

Team Members

- Stefano Buglione, SAB21S
- Lauryn Brown, LB16E
- Marie Schwarz, MCS23G
- Nicolas Fernandez, NGF22

Project Description

- Our team (called TheParents) plan to create an interactive AI model that focuses on the topics in personalized medicine.
- The AI will be called AMI (Artificial Medical Intelligence).
- Our goal is to create a functional LLM that provides source transparency.
- The AI will respond to a given prompt and provide the list of sources that were used in the response. It will provide a citation and potential link to the source

Assignment of Responsibilities for this Week

- **Stefano Buglione:**
 - Design source transparency and citation system
- **Lauryn Brown:**
 - Design AML avatar appearance and behavior
- **Marie Schwarz:**
 - System architecture design
 - LLM training pipeline and deployment design
- **Nicolas Fernandez:**
 - UI/UX wireframes

Current state of the project

- Up-to date Gantt Chart:

Tasks	Assigned To	Start Date	Due Date	Status	Completion %
Team Recruitment	-	01/05/26	01/07/26	Complete	100%
Project Idea	-	01/05/26	01/12/26	Complete	100%
Project Prospectus	-	01/12/26	01/23/26	Complete	100%
Group Meetings	-	Saturday	8PM EST	In Progress	60%
Define system scope and constraints	-	1/12/2026	1/30/2026	Complete	100%
Identify user roles and use cases	Nicolas Fernandez	2/1/2026	2/8/2026	Complete	100%
Formalize functional requirements	Stefano Buglione	1/30/2026	2/5/2026	Complete	100%
Non-functional requirements(LLM, ethics, and privacy laws)	Laurn Brown	2/4/2026	2/9/2026	Complete	100%
Finalize initial SRS (Software Requirements Specification)	Marie Schwarz	2/6/2026	2/14/2026	Complete	100%
Define AI avatar requirements	Stefano Buglione	2/6/2026	2/14/2026	Complete	100%
Analyze existing AI medical systems	Nicolas Fernandez	2/15/2026	2/20/2026	Complete	100%
Dataset and source evaluation	Stefano Buglione	2/15/2026	2/25/2026	Complete	100%
Analyze bias and hallucination risks	Marie Schwarz	2/18/2026	2/28/2026	Complete	100%
Feasibility for avatar tools analysis	Laurn Brown	2/20/2026	3/1/2026	Complete	100%
Finalize system requirements	Marie Schwarz	2/20/2026	3/5/2026	Complete	100%
System architecture design	Marie Schwarz	3/6/2026	3/12/2026	Complete	100%
LLM training pipeline and deployment design	Marie Schwarz	3/6/2026	3/18/2026	Complete	100%
UI/UX wireframes	Nicolas Fernandez	3/10/2026	3/20/2026	In Progress	0%
Design AMI avatar appearance and behavior	Laurn Brown	3/6/2026	3/22/2026	In Progress	0%
Design source transparency and citation system	Stefano Buglione	3/15/2026	3/25/2026	In Progress	33%
Set up chatbot framework	-	3/21/2026	3/28/2026	Not Started	0%
Implement response generation	-	3/25/2026	4/2/2026	Not Started	0%
Integrate APIs	-	3/28/2026	4/5/2026	Not Started	0%
Implement text-to-speech	-	3/30/2026	4/3/2026	Not Started	0%
Frontend implementation	-	4/1/2026	4/5/2026	Not Started	0%

Current State of the Project

- Marie's work for:
 - System architecture design

1. Architecture Goals and Constraints

- Provide educational answers about personalized medicine without offering diagnosis or treatment.
- Ensure transparency through citations from validated and peer-reviewed sources.
- Reduce hallucinations and bias using Retrieval-Augmented Generation (RAG), guardrails, and disclaimers.
- Provide a web-based application with avatar visualization, text responses, and optional text-to-speech.

2. High-Level System Architecture

- Client Layer: Web UI with chat interface, avatar renderer, audio controls, and optional speech-to-text.
- Backend Layer: API services for chat processing, guardrails, retrieval, LLM generation, citations, and feedback.
- Knowledge Layer: Curated medical corpus stored in document storage and vector databases for semantic search.
- Observability Layer: Logging, metrics, uptime monitoring, and evaluation pipelines.

3. Main System Components

- Frontend (Browser): Chat interface, avatar state controller (Idle, Loading, Speaking), audio playback module, feedback widget.
- Backend Server: API Gateway, conversation orchestrator, policy and guardrails engine, retrieval service, LLM generator, citation formatter, text-to-speech service.
- Data Stores: Vector database for embeddings, document store for source material, feedback database, and system logs.

Current State of the Project

- Marie's work for:
 - System architecture design (cont.)

4. Data Flow

- 1. User submits a prompt through the web chat interface.
- 2. The frontend sends the request to the backend API.
- 3. Guardrails check if the prompt violates scope (diagnosis/treatment).
- 4. Retrieval service searches peer-reviewed knowledge sources using embeddings.
- 5. LLM generates a response grounded in retrieved information.
- 6. Citation service attaches relevant source references.
- 7. Response is returned to the UI and optionally converted to audio through the TTS service.
- 8. User may submit feedback after the interaction.

5. Deployment Architecture

- Browser client hosting the user interface.
- Backend API server handling orchestration and AI logic.
- Vector database for semantic retrieval.
- Document storage for validated medical sources.
- Feedback database for user interaction data.

6. Security, Privacy, and Safety Controls

- Minimal storage of user data with anonymized session IDs.
- Encryption and redaction if sensitive data is stored.
- Guardrails to prevent medical diagnosis or treatment recommendations.
- Use of validated sources to minimize hallucination risks.

7. Requirement Traceability

- Chat interaction requirement => Chat UI and Conversation Orchestrator.
- Educational-only responses => Policy and Guardrails Engine.
- Citations requirement => Retrieval and Citation Services.
- Avatar and audio interaction => Avatar Controller and TTS Service.
- Performance requirements => Monitoring and scalable backend architecture.

Current State of the Project

- Marie's work for:
 - LLM training pipeline and deployment design

1. Design Objectives

- Ensure responses are grounded in validated and peer-reviewed medical sources with visible citations.
- Maintain educational-only responses without diagnosis or treatment recommendations.
- Reduce hallucinations and bias using Retrieval-Augmented Generation (RAG), guardrails, and dataset filtering.
- Meet performance requirements such as response time under 30 seconds and system uptime above 95%.

2. Training Pipeline Overview

- Data ingestion and governance for collecting and validating trusted medical datasets.
- Dataset preparation separating knowledge sources and behavioral training examples.
- Optional supervised fine-tuning focused on safety, refusal behavior, and response formatting.
- Evaluation pipelines to measure groundedness, bias risk, hallucination rates, and latency.

3. Data Ingestion and Governance

- Register data sources and maintain metadata including origin, license, and reliability.
- Apply de-identification and privacy checks to prevent storing sensitive information.
- Filter low-quality or non-cited conversational data that may increase hallucination risk.
- Tag datasets for known bias risks such as geographic or demographic imbalance.

4. Dataset Preparation

- Knowledge Track: convert trusted sources into text chunks and generate embeddings for vector search.
- Store document metadata such as title, authors, year, and publication source.
- Behavior Track: create examples for refusal responses, disclaimers, and citation formatting.
- Train the model to ask clarifying questions when prompts are ambiguous.

5. Model Training Strategy

- Use a pretrained large language model as the base model.
- Apply light supervised fine-tuning to reinforce safe and consistent response behavior.
- Use RAG as the primary mechanism for factual medical knowledge retrieval.
- Require the model to ground responses in retrieved passages before generating answers.

Current State of the Project

- Marie's work for:
 - LLM training pipeline and deployment design (cont.)

6. Evaluation and Validation

- Test groundedness by verifying responses match retrieved knowledge sources.
- Evaluate hallucination risk using prompts designed to test unsupported claims.
- Assess bias by testing responses across different populations or contexts.
- Measure performance against system latency and accuracy thresholds before release.

7. Deployment Architecture

- Frontend web application providing the chat interface and avatar visualization.
- Backend API services handling chat requests, guardrails, retrieval, generation, and feedback.
- Retrieval service using embeddings and vector search for knowledge retrieval.
- Text-to-speech service for optional audio output.

8. Data Storage Components

- Vector database for embeddings and semantic search.
- Document storage for validated medical source content.
- Feedback database for user interaction ratings and comments.
- System logs and metrics for monitoring performance and reliability.

9. Runtime Request Flow

- User submits a prompt through the web interface.
- The backend processes the request and checks guardrails for scope compliance.
- The retrieval service finds relevant medical knowledge sources.
- The LLM generates an answer grounded in retrieved information.
- Citations are appended to the response and returned to the user.
- Optional audio output is generated through the text-to-speech service.
- User feedback is collected for continuous improvement.

10. Monitoring and Continuous Improvement

- Monitor hallucination rates, refusal accuracy, and citation coverage.
- Track user feedback to identify improvement opportunities.
- Update the retrieval knowledge base with new peer-reviewed sources.
- Run evaluation tests before deploying updated models.

Planned Activities for the following week(s)

- Stefano Buglione:
 - Finalize: Design source transparency and citation system
- Lauryn Brown:
 - Finalize: Design AMI avatar appearance and behavior
- Nicolas Fernandez:
 - Finalize: UI/UX wireframes
 - Acquiring additional test data (using PRISMA)

Issues Not Resolved

- There have not been any major issues in the current state of development
- The team is still discussing AMI's final appearance and behavior, which will be solved by next week.